EXCEL REFERENCING & LOGICAL FUNCTIONS

Complete learning handbook | Lessons 47-51

Relative, absolute and mixed references • Named ranges • IF, AND, OR and comparison operators

Included: Definitions, Excel navigation, explained formulas, step-by-step instructions, reproducible datasets, expected outputs, important limitations and quick-reference tables.

Excluded: Practice assignments and exercise questions.

Lesson	Main topics
47	Cell references, relative referencing, copy and fill, moving ranges
48	Absolute and mixed references, tax and discount formulas, F4 shortcut
49	Named ranges, Name Box, Define Name, Name Manager and limitations
50	IF function, text conditions, numerical comparisons and budget checks
51	AND and OR, truth tables, date conditions, nested DATE and IF

Examples use illustrative fruit, product, and transaction data. Formulas assume comma-separated arguments and standard desktop Microsoft Excel.

Lesson 47 | Relative Referencing

1. Cell references and why they matter

A cell reference points to an existing cell value. A formula can use changing worksheet values without retyping numbers. For example, if B2 contains 7 and C2 contains 10:

```
=B2*C2 -> 70
```

```
=PRODUCT(B2,C2) -> 70
```

The PRODUCT function multiplies its numeric arguments; both examples calculate the same result.

2. Relative referencing

Relative references are Excel's default. When a formula is copied, referenced row and column positions shift by the same displacement as the formula.

Destination	Formula after copying	Reason
D2	=B2*C2	Original formula
D3	=B3*C3	Copied down one row
D4	=B4*C4	Copied down two rows
E2	=C2*D2	Copied right one column

3. Fruit purchase dataset

Enter the following values in cells A1:D6 of a worksheet named 47_RelativeReferencing. Column D is calculated.

Row	A - Fruit	B - Price/unit	C - Quantity	D - Total
2	Apple	7	10	70
3	Mango	12	5	60
4	Orange	8	8	64
5	Banana	5	12	60
6	Grapes	15	4	60

4. Enter and copy the formula

1. Select D2, type =B2*C2 and press Enter. Result: 70.
2. Copy D2 using Ctrl+C and paste into D3 using Ctrl+V. The formula becomes =B3*C3.
3. To fill all rows, drag the bottom-right fill handle of D2 through D6, or select D2:D6 and press Ctrl+D.

5. Relative ranges

Both ends of a range shift when the formula is copied. For example, moving a formula down two rows changes:

```
=SUM(B2:B5) -> =SUM(B4:B7)
```

> Important: Relative references shift when copying or filling formulas. They do not mean that existing values are hardcoded or that a reference will always move only downward.

6. Key distinctions

Expression	Effect
=7*10	Fixed numbers: does not use worksheet input cells
=B2*C2	Calculates from worksheet cells; updates if input changes
Copying D2 to D3	B2 -> B3 and C2 -> C3
Copying D2 to E2	B2 -> C2 and C2 -> D2

Lesson 48 | Absolute and Mixed Referencing

1. Three reference types

Type	Example	Copied down one row	Copied right one column
Relative	A1	A2	B1
Absolute	$A$1	$A$1	$A$1
Mixed - fixed column	$A1	$A2	$A1
Mixed - fixed row	A$1	A$1	B$1

The dollar sign ($) fixes the component immediately following it during copy/fill. $A$1 fixes both parts, $A1 fixes the column, and A$1 fixes the row.

2. Revenue and fixed tax-rate example

Create worksheet 48_AbsoluteMixedRef, with data in A1:F5. Put the tax rate in H1 = 18%.

Row	Product	Quantity (B)	Unit price (C)	Revenue (D)	Tax (E)
2	Pizza	5	200	1,000	180
3	Burger	3	150	450	81
4	Pasta	4	250	1,000	180
5	Sandwich	6	100	600	108

1. In D2, enter =B2*C2; fill down to D5.
2. In E2, enter =D2*$H$1; fill down to E5.
3. Verify that D2 becomes D3, D4, etc., while $H$1 stays fixed.

```
E2 =D2*$H$1 E3 =D3*$H$1 E4 =D4*$H$1
```

Important: Incorrect formula =D2*H1 shifts H1 to H2 when copied down, so the tax reference no longer points to the rate.

3. Fixed discount rate

Enter H2 = 12.5%. Calculate the final price after discount in column F:

```
F2 =D2-(D2*$H$2)
```

Equivalent: `=D2*(1-$H$2)`

Product	Revenue	Discount rate	Final price
Pizza	1,000	12.5%	875
Burger	450	12.5%	393.75
Pasta	1,000	12.5%	875
Sandwich	600	12.5%	525

4. Mixed referencing in a two-way calculation

Put column headers B10=2, C10=3, D10=4 and row headers A11=5, A12=10, A13=15. Type the formula in B11, then copy across and down:

```
=$A11*B$10
```

Row input	2	3	4
5	10	15	20
10	20	30	40
15	30	45	60

The column A reference stays on A as the formula moves right. Row 10 stays fixed as the formula moves down.

5. F4 reference shortcut

While editing a formula, position the cursor within a cell reference and press F4 to cycle:

```
B2 -> $B$2 -> B$2 -> $B2 -> B2
```

On some laptops use Fn+F4. The shortcut applies while editing a formula reference, not merely after selecting a cell.

> Important: Absolute/mixed references constrain the position referenced during copying; they do not lock the input cell against editing.

Lesson 49 | Named Ranges

1. What a named range does

A named range assigns a reusable name to one cell, part of a row/column, an entire column, or a larger dataset. For example, Vendor_List can refer to B2:B6.

```
=COUNTA(B2:B6) and =COUNTA(Vendor_List)
```

These return the same count when Vendor_List refers to B2:B6.

2. Fruit/vendor dataset

In a worksheet named 49_NamedRanges, enter:

Row	A - Fruit	B - Vendor	C - Price	D - Month
2	Apple	FreshMart	120	January

Row	A - Fruit	B - Vendor	C - Price	D - Month
3	Mango	GreenFarm	150	February
4	Orange	FreshMart	80	March
5	Banana	NatureFresh	50	April
6	Grapes	GreenFarm	100	May

3. Create names - three Excel methods

Method	Desktop Excel steps
Name Box	Select B2:B6 -> click Name Box left of formula bar -> type Vendor_List -> press Enter.
Define Name	Select A2:D6 -> Formulas > Define Name -> enter Fruit_Data -> verify Refers to -> OK.
Name Manager	Formulas > Name Manager > New -> enter Fruit_Prices -> set Refers to C2:C6 -> OK.

The workbook-level references in this example are:

```
Vendor_List = '49_NamedRanges'!$B$2:$B$6
Fruit_Data = '49_NamedRanges'!$A$2:$D$6
Fruit_Prices = '49_NamedRanges'!$C$2:$C$6
```

4. Select and use a name

1. Use the Name Box dropdown to select Vendor_List; Excel selects its cells.
2. Type =COUNTA(Vendor_List) to count populated vendor entries: 5.
3. Use Fruit_Prices with the basic statistical functions below.

Formula	Expected output
=SUM(Fruit_Prices)	500
=AVERAGE(Fruit_Prices)	100
=COUNT(Fruit_Prices)	5
=MAX(Fruit_Prices)	150
=MIN(Fruit_Prices)	50
=COUNTA(Fruit_Data)	20

> Important: COUNTA counts entries, not unique values. The five vendor entries represent three distinct vendor names.

5. Edit, delete and scope

Edit: Formulas > Name Manager > select the name > Edit > update Refers to (for example, B2:B8) > OK. Delete: Name Manager > select > Delete > confirm. Deleting a name does not delete its data, but formulas using it may show #NAME?.

Scope: Workbook-scoped names can be used across sheets; worksheet-scoped names belong to a particular sheet. From another sheet, a sheet-local name can be qualified, for example:

```
=A2*Sheet1!Tax_Rate
```

6. Naming rules and expanding data

Valid names	Invalid / problematic names
Vendor_List, Fruit_Prices, Tax_Rate	Vendor List (space); 1Vendor (starts with number)
Monthly_Sales	A1 (cell-reference style); Sales-Amount (hyphen)
_Sales	R or C by themselves (reserved)

Named ranges are not necessarily dynamic. A fixed B2:B6 reference does not automatically include a new entry in B7. Update it in Name Manager or use an Excel Table for a growing list (select data > Insert > Table). Names are case-insensitive within their scope.

Lesson 50 | IF Function

1. Purpose and syntax

IF tests one logical expression and returns an assigned result for TRUE or FALSE.

```
=IF(logical_test,value_if_true,value_if_false)
```

Argument	Meaning
logical_test	Condition such as B2="Fantastic" or F2>6
value_if_true	What Excel returns when condition is met
value_if_false	What Excel returns when condition is not met

2. Text conditions - party decision

In 50_IFFunction, enter the following data in columns A:C. In C2, type the formula and fill down:

```
=IF(B2="Fantastic","Will Go","Will Not Go")
```

Party	Food rating (B)	Decision (C)
A	Good	Will Not Go
B	Fantastic	Will Go
C	Okay	Will Not Go
D	Fantastic	Will Go
E	Poor	Will Not Go

Excel tests the text in B2. Another scenario from the lecture checks dress quality:

```
=IF(B2="Spectacular","Will Go","Will Not Go")
```

> Important: Text literals and text output labels belong in double quotation marks. References and numeric values normally do not.

3. Budget classification - $6 limit

Use a separate fruit-price table in columns E:G; prices are in F2:F6. Place the following in G2 and fill down:

```
=IF(F2>6,"Over Budget","Within Budget")
```

Fruit	Purchase price	Output
Apple	5.00	Within Budget
Mango	8.50	Over Budget
Orange	6.00	Within Budget
Banana	3.50	Within Budget
Grapes	6.10	Over Budget

A price of exactly $6 is within the budget because 6>6 is FALSE. Using >= instead would classify $6 as over budget.

```
=IF(F2>=6,"Over Budget","Within Budget")
```

4. Reference the budget value rather than hardcoding it

Store the budget 6 in K1. Use an absolute reference so the same budget cell applies as the formula fills down:

```
=IF(F2>$K$1,"Over Budget","Within Budget")
```

Changing K1 recalculates the output. IF can also return numbers, such as =IF(A2>=50,1,0).

5. Common IF errors

Issue	Check
Text not in double quotes	Use "Pass", "Fail", "Fantastic", etc.
Wrong boundary operator	> excludes equality; >= includes it
Unexpected results after fill	Check relative vs absolute cell references
Syntax error	Ensure commas/semicolons and closing parentheses are correct

Lesson 51 | AND, OR & Logical Operators

1. Multiple logical conditions

AND and OR combine multiple logical expressions. They return TRUE/FALSE; IF can wrap them to return custom labels.

```
=AND(logical1,[logical2],...)
```

```
=OR(logical1,[logical2],...)
```

Condition A	Condition B	AND result	OR result
TRUE	TRUE	TRUE	TRUE
TRUE	FALSE	FALSE	TRUE

Condition A	Condition B	AND result	OR result
FALSE	TRUE	FALSE	TRUE
FALSE	FALSE	FALSE	FALSE

2. Transaction dataset

Create worksheet 51_AND_OR, with the following columns A:E. Ensure purchase dates are stored as real Excel dates.

Product	Purchase (B)	Sales (C)	Country (D)	Date (E)
Laptop	43,000	44,000	United States	19-Mar-2026
Mobile	45,000	44,000	United States	08-Apr-2026
Tablet	42,000	46,000	France	24-Apr-2026
Monitor	47,000	44,000	India	05-May-2026
Keyboard	40,000	45,000	India	10-May-2026
Mouse	38,000	42,000	United States	15-May-2026

3. AND: numerical + text conditions

Check that sales exceed purchase amount and country is United States. In F2, fill down:

```
=AND(C2>B2,D2="United States")
```

Product	AND result	Reason
Laptop	TRUE	Sales exceed cost and country matches
Mobile	FALSE	Sales do not exceed cost
Tablet	FALSE	Country does not match
Monitor	FALSE	Both conditions fail
Keyboard	FALSE	Country does not match
Mouse	TRUE	Both conditions pass

Another lecture condition: purchase amount at least 44,000 and purchased in India:

```
=AND(B2>=44000,D2="India")
```

Only Monitor qualifies in the illustrated dataset.

4. OR: either condition can qualify

Check that sales exceed cost or country is United States:

```
=OR(C2>B2,D2="United States")
```

In the dataset, all products return TRUE except Monitor. Another condition is purchase amount below 45,000 OR country India:

```
=OR(B2<45000,D2="India")
```

This returns FALSE only for Mobile in the illustrated dataset. The 45,000 boundary is excluded by <.

5. Operators for logical tests

Operator	Meaning	Illustration
=	Equals	=10=10 returns TRUE
>	Greater than	=10>5 returns TRUE
<	Less than	=10<5 returns FALSE
>=	Greater than or equal to	=10>=10 returns TRUE
<=	Less than or equal to	=10<=10 returns TRUE
<>	Not equal to	=10<>5 returns TRUE

6. DATE function in logical conditions

Excel compares actual date serial values, not human-readable date strings. Quoted strings such as "01-05-2026" may be interpreted inconsistently. Construct the date explicitly:

```
=DATE(2026,5,1) -> 01-May-2026 (formatted as date)
```

Check purchase date on/after 1 May 2026 and cost below 44,000:

```
=AND(E2>=DATE(2026,5,1),B2<44000)
```

Expected TRUE: Keyboard, Mouse. Other products return FALSE.

Check cost at most 45,000 or purchase date on/after 1 May 2026:

```
=OR(B2<=45000,E2>=DATE(2026,5,1))
```

Expected result: TRUE for all six products, since every record satisfies at least one test.

> Important: Use the full year 2026 in DATE(2026,5,1). DATE(26,5,1) does not mean 2026 in Excel.

7. Nested functions: IF with AND or OR

The lecture introduces IF and AND/OR independently; combining them is a useful extension for descriptive results:

```
=IF(AND(C2>B2,D2="United States"),"Good Business","Not Qualified")
```

Good Business: Laptop, Mouse. All others: Not Qualified.

```
=IF(OR(B2<45000,D2="India"),"Eligible","Not Eligible")
```

Only Mobile returns Not Eligible in this dataset.

8. Mistakes and limits

Situation	Correct practice
AND returns FALSE unexpectedly	Check whether every required condition passes
OR returns TRUE unexpectedly	Only one condition must pass
Date comparison inconsistent	Confirm date values are real dates; use DATE(year,month,day)
Equality boundaries incorrect	Check < vs <= and > vs >=
Errors in input cells	Investigate source errors before trusting logical results
Many conditions	AND and OR support up to 255 logical arguments in current Excel

Consolidated Formula Reference

Goal	Formula
Relative multiplication	=B2*C2
PRODUCT alternative	=PRODUCT(B2,C2)
Fixed tax	=D2*$H$1
Final price after fixed discount	=D2*(1-$H$2)
Mixed two-dimensional multiplication	=$A11*B$10
Count a named vendor range	=COUNTA(Vendor_List)
Sum a named price range	=SUM(Fruit_Prices)
Text-based IF	=IF(B2="Fantastic","Will Go","Will Not Go")
Budget check	=IF(F2>6,"Over Budget","Within Budget")
Two mandatory conditions	=AND(C2>B2,D2="United States")
At least one condition	=OR(C2>B2,D2="United States")
Date + cost condition	=AND(E2>=DATE(2026,5,1),B2<44000)
Custom result from AND	=IF(AND(C2>B2,D2="United States"),"Good Business","Not Qualified")

Final checklist

Confirm the correct reference type before copying; keep shared rates fixed; review named-range definitions after data changes; use quoted text correctly; distinguish > from >=; choose AND only when every condition is mandatory; choose OR for alternatives; and represent dates with real date values or DATE().

End of Lessons 47-51. This handbook contains reference explanations, procedures, examples and expected results, with no practice assignments.